A Cross-Level Spectral–Spatial Joint Encode Learning Framework for Imbalanced Hyperspectral Image Classification

Dabing Yu, Qingwu Li, Xiaolin Wang, Chang Xu, and Yaqin Zhou

***Abstract*— Convolutional neural networks (CNNs) have dominated the research of hyperspectral image (HSI) classification, attributing to the superior feature representation capacity. Fast patch-free global learning (FPGA) as a fast learning framework for HSI classification has received wide interest. Despite their promising results from the perspective of fast inference, recent works have difficulty modeling spectral–spatial relationships with imbalanced samples. In this article, we revisit the encoder–decoder-based fully convolutional network (FCN) and propose a cross-level spectral–spatial joint encoding (CLSJE) framework for imbalanced HSI classification. First, a multiscale input encoder and multiple-to-one multiscale features connection are introduced to obtain abundant features and facilitate multiscale contextual information flow between the encoder and the decoder. Second, in the encoder layer, we propose the spectral–spatial joint attention (SSJA) mechanism consisting of high-frequency spatial attention (HFSA) and spectral-transform channel attention (STCA). HFSA and STCA encode spectral–spatial features jointly to improve the learning of the discriminative spectral–spatial features. Powered by these two components, CLSJE enjoys a high capability to capture both spatial and spectral dependencies for HSI classification. Besides, a class-proportion sampling strategy is developed to increase the attention to insufficiency samples. Extensive experiments demonstrate the superiority of our proposed CLSJE both at classification accuracy and inference speed, and show the state-of-the-art results on four benchmark datasets. Code can be obtained at https://github.com/yudadabing/CLSJE.**

***Index Terms*— Cross-level feature aggregation (CLA), hyperspectral image (HSI) classification, imbalanced samples, inverse-weighted loss, spectral–spatial joint attention (SSJA).**

NOMENCLATURE

HSI	Hyperspectral image.
HF	High-frequency.
CNN	Convolutional neural network.
FCN	Fully convolutional network.
CLSJE	Cross-level spectral-spatial joint encoding.
SSJA	Spectral–spatial joint attention.
SFFM	Supervised feature fusion module.
HFSA	High-frequency spatial attention.
STCA	Spectral-transform channel attention.
CSS	Class-proportion sampling strategy.
DCT	Discrete cosine transform.

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I. INTRODUCTION

HYPERSPECTRAL imaging obtains abundant spectral information by capturing hundreds of continuous wavebands ranging from the visible spectrum, near-infrared to far-infrared [1], [2], [3]. Benefiting from the high spectral resolution, HSIs yield more discriminating feature representations than traditional RGB images. At its core, HSI classification assigns a unique semantic label to each hyperspectral pixel [4], [5], [6]. HSI classification plays an essential role in remote sensing scenes, such as environmental monitoring, urban planning, and geological exploration [7], [8]. However, HSI classification is highly challenging due to the Hughes phenomenon [9] caused by the high-dimensional data redundancy and the limited labeled HSI samples.

Early works exploit the spectral curves of different ground objects to classify HSIs by kernel classifier-based methods [10] or feature representation-based ones [11] while paying less attention to the spatial information of HSIs. Their performance is limited due to spectral noise like highlight and shadow. Besides, the classification accuracy of the above algorithms decreases sharply as the number of spectral bands increases to the threshold. Spatial and spectral features have been exploited together to achieve a more accurate HSI classification [12], [13], [14], [15], [16]. Spectral–spatial classification methods, mainly including Markov random field (MRF) [13], [14], extended morphological profiles (EMPs) [15], [16], gray level co-occurrence matrix [17], and the Gabor filter [18], have been proposed. However, all the spectral–spatial parameters are given manually without further explanation and are established on broad prior knowledge and empirical data [19].

Recent years have witnessed the rapid advancement of HSI classification motivated by the widely successful deep learning technology. CNNs, as hierarchical data-driven learning frameworks, can adaptively extract the spectral–spatial features of HSIs and achieve end-to-end training. CNN-based methods [20], [21], [22] significantly boost the accuracy compared with the traditional methods. Numerous variants of CNNs, e.g.,

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3-D-CNN [23], multichannel CNN [24], recurrent neural networks [25], convolutional recurrent neural networks [26], [27], and other novel networks [28], have been developed to improve the learning ability of spectral–spatial features. Most above CNN-based models follow the patch-based local learning frameworks. However, patch-based local learning frameworks exhibit two major flaws. On the one hand, patch-based frameworks have a high computational complexity due to the large overlapped areas among adjacent patches. On the other hand, these models are difficult in modeling long-term dependencies without adaptive spatial context.

To address this challenge, a fast patch-free global learning (FPGA) framework [29] was proposed to reduce redundant calculations and model the long-range spatial features. Formulated within the patch-free learning framework, a global end-to-end CNN is proposed to make maximum exploitation of the global spatial information. FPGA makes an assumption that the labeled samples per class are the same in each hierarchical training dataset. However, this assumption obtains insufficient performance on datasets with insufficient and imbalanced samples (i.e., the Indian Pines (IP) dataset). Zhu *et al.* [30] developed a spectral–spatial dependent global learning framework according to the FPGA framework, which further handled the sample problem of insufficiency and imbalance.

However, classification accuracy is still limited. The main reasons are twofold. First, the standard U-Net is to leverage high-level semantics by upsampling low-resolution feature maps and fusing them with high-resolution feature maps. However, only the coarse-to-fine feature flow results in inefficient feature aggregation for limited samples. Meanwhile, the significant spatial features may be lost in the downsampled modules and pooling layer. Second, redundant HSI data raise the difficulty in the HF feature extraction. CNN can view the data at a much higher granularity than human can. It tends first to pick up the low-frequency component and then gradually picks up the HF component to achieve higher accuracy [31]. For coarse-grained classification tasks, it is apparent that traditional CNN with numerous low-frequency components and few HF components is sufficient. The hyperspectral classification task aims to classify various urban land-cover/land-use classes. It contains coarse-grained classification (e.g., roads and water) and fine-grained classification (e.g., healthy grass and stressed grass). HF components, e.g., local textures, are discriminative and vital for fine-grained classification. Especially for limited training samples and redundant HSI data, the HF feature denoted by the discriminative feature is valuable and hard to extract.

Motivated by these issues, in this article, we propose a cross-level spectral–spatial joint encode learning framework called CLSJE. First, our CLSJE equips the multi-input encoder and multiple-to-one lateral connections. Scaled input images are added into corresponding encoder layers as extra inputs to counter the information loss from the pooling layer and downsampled operation. Meanwhile, the SFFM is proposed to integrate both the downsampled feature extracted and the scaled input in a supervised manner. The multiple-to-one lateral connections aggregate features at higher, current, and lower levels. In this way, context information is shared and aggregated in each scale, thus encouraging the information flowing to the desired direction. Second, the SSJA mechanism combines HFSA and STCA, which is proposed to extract the HF spatial features and spectral features of HSIs. In addition, a CSS is developed to obtain a balance minibatch per class and increase the attention to hard-to-classify samples.

The main contributions of this article are summarized as follows.

1) A novel CLSJE framework is proposed for HSI classification. The CLSJE can achieve faster learning from the global spectral–spatial information and better accuracy for imbalanced HSI classification. We conduct extensive experiments to confirm that our model outperforms the state-of-the-art methods.
2) A multiscale input encoder and the multiple-to-one lateral connection are proposed to prevent the feature missing by downsampled operation, boost information flow flexibility, and achieve an asymmetric combination of multiscale spectral–spatial features.
3) To extract the detailed spectral–spatial features, SFFM and SSJA are embedded in the encoder layer. SFFM aims to fuse the downsampled HSI and decoded features in a supervised way, which takes advantage of the complementary information from encoded features. SSJA is developed to extract the discriminative feature representations, which tends to avoid aliasing, preserve useful spatial information, and adaptively recalibrate spectral feature responses at different frequencies.
4) To focus more on hard-to-classify samples and guarantee the convergence rate of training, a CSS with the inverse-weighted cross-entropy loss is designed to assist with training the proposed method CLSJE. Benefiting from the proposed sampling strategy, the weight of easy-to-classify samples is reduced, and the model convergence is sped up.

This article is organized as follows. In Section II, we outline the related work in the field of HSI classification. Following that, the proposed CLSJE framework is described in detail in Section III. Experimental results are presented in Section IV, and ablation studies are discussed in Section V. The conclusions are given in Section VI. Abbreviations used in this article are summarized in Nomenclature.

II. Related Work

In this section, a thorough review of CNN-based methods for HSI classification is outlined due to the ability to adaptively extract spectral–spatial features.

Generally, CNN-based methods can transform images into more recognizable knowledge by automatically extracting abstract features from the bottom to the top level of semantics [19], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41]. Hu *et al.* [33] used 1-D-CNN for HSI pixel-level classification, demonstrating that 1-D-CNN is practical for remotely sensed HSI classification. 2-D-CNN was proposed in [38] to perform an image-level HSI classification by treating a neighborhood window around each labeled pixel as a training sample. To exploit the spectral–spatial information in HSIs and

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simplify preprocessing, Hamida *et al.* [23] developed a 3-D-CNN and input the 3-D cube in a small size to process joint spectral–spatial information and improved the classification performance. A spectral–spatial residual network (SSRN) was proposed in [40], in which the cascading spatial and spectral residual blocks were used to extract features from abundant spatial contexts and spectral properties. Subsequently, Chen *et al.* [24] proposed a multichannel CNN which merged 1-D-CNN, 2-D-CNN, and 3-D-CNN and solved the drawbacks of each feature extraction channel. Wu and Prasad [26] proposed a convolutional recurrent neural network to model the dependencies between different spectral bands and learn more discriminative feature representations for HSI classification. However, its main drawback is that the network training depends on sufficient labeled samples, and the spectral–spatial information of unlabeled samples is ignored.

To handle the high cost of obtaining sufficient labeled samples, several works used patches composed of few labeled samples and substantial unlabeled samples of HSI for training. Chen *et al.* [32] utilized a stacked autoencoder to merge spatial-dominated information and spectral representation, which achieved accurate classification results. The work by Mnih *et al.* [42] was dedicated to performing enhanced learning in the high-dimensional input situation and propagating HSI labels from annotated pixels to unannotated ones by adopting a multilayer neural network. Pan *et al.* [43] proposed a multigrained network for HSI classification by learning convolution kernels with all unlabeled samples in a semisupervised manner. Fang *et al.* [8] designed a lightweight 3-D-CNN and generated pseudolabels by approximated rank-order clustering to make full use of large unlabeled HSI samples. Mei *et al.* [44] developed a 3-D convolutional autoencoder to train all the parameters involved in the network without labeled training samples, which achieved accurate classification in an unsupervised mode. In addition, a generative adversarial network was addressed in [45] to relieve the negative impact of limited training samples by learning real HSI data distribution. Jia *et al.* [46] proposed a superpixel-based CNN to achieve the internal consistency of each patch and established a patch-based similarity matrix, which showed the superiority of HSI classification performance. Despite the solution of insufficient samples to some extent, the scattered and irregular materials in HSIs are likely to be incorrectly segmented by patches, leading to high computational complexity and even modeling failure for the HSI classification network.

To deal with the issues mentioned above, more individualized sampling strategies and learning frameworks are needed [47]. FCN formulated within the patch-free learning framework was first proposed for image segmentation and achieved great success [48]. Given the sparsity of HSI data that HSI samples only contain a discrete set of labeled pixels rather than a set of labeled images, Xu *et al.* [49] proposed a spectral–spatial FCN for HSI classification with an end-to-end, pixel-to-pixel architecture, which avoided patch extraction. A novel mask matrix and spectral–spatial weighting factors were further developed to assist the back-propagation and adaptively learn both features in the training stage. An FPGA framework [29] was explored to avoid redundant computation and guarantee model convergence. In FPGA, FCN with an encoder–decoder architecture was utilized to mine the global spatial context information of the whole image and achieved fast inference. However, the long-tail distribution of HSI datasets poses a tremendous challenge for classification performance. Rectifying the decision boundaries on the imbalanced classes by fine-tuning and optimizing parameters via loss reweighting strategies are the common technique for solving class-imbalance issue [50], [51], [52]. Zhu *et al.* [30] proposed a global convolutional long short-term memory network and sampling strategy to classify insufficient and imbalanced HSI data. The problems of sample imbalance and model convergence can be mitigated to a certain degree. These patch-free-based methods only utilized one-to-one lateral connections and single-scale input, meaning that the significant spectral–spatial features may be lost in the concatenated network. There is still a challenge in extracting spectral–spatial features from imbalanced HSI samples and achieving fast and accurate HSI classification.

In pursuit of overcoming the above issues, we construct a CLSJE learning framework that leverages cross-scale feature flexible flowing and effective spatial–spectral features extraction. The SSJA mechanism activates the discriminative spatial–spectral features by individually encoding spectral and spatial information, and cross-branch feature aggregation. An SFFM is applied to integrate the asymmetrical multiscale spectral–spatial features. Moreover, we also embed a CSS to achieve an accurate imbalanced HSI classification.

III. Proposed Method

In this section, we introduce CLSJE that fully exploits multiscale spectral and spatial features for HSI classification. The overall architecture of CLSJE is shown in Fig. 1. Our CLSJE is an encoder–decoder FCN with significant modifications to suit HSI classification. Inspired by the coarse-to-fine spatial image pyramid [53], we introduce downsampled HSIs to each encoder layer, which provides complementary spatial information with a bigger receptive field. Correspondingly, the decoder layer aggregates features from all encoder layers and decoded features from the higher level decoding layers. In each encoder layer, SFFM is first built to fuse the downsampled HSI and the downsampled features from the encoder layer. Importantly, the spectral and spatial features are then effectively extracted using SSJA. Sections III-A–II-D detail the proposed CLSJE, including cross-level feature aggregation (CLA) architecture, SSJA, SFFM, and the sampling strategy.

A. Cross-Level Feature Aggregation Architecture

1) Multi-Input Encoder: The use of multiscale images as inputs for FCN has proven to be effective in computer vision tasks, such as depth map super-resolution [54], object detection [53], and image deblurring [55]. Different from conventional FCN, in our CLSJE, each encoder layer takes an extra scale HSI as input. In other words, expect the first encode layer, each encoder layer fuses the decoded features

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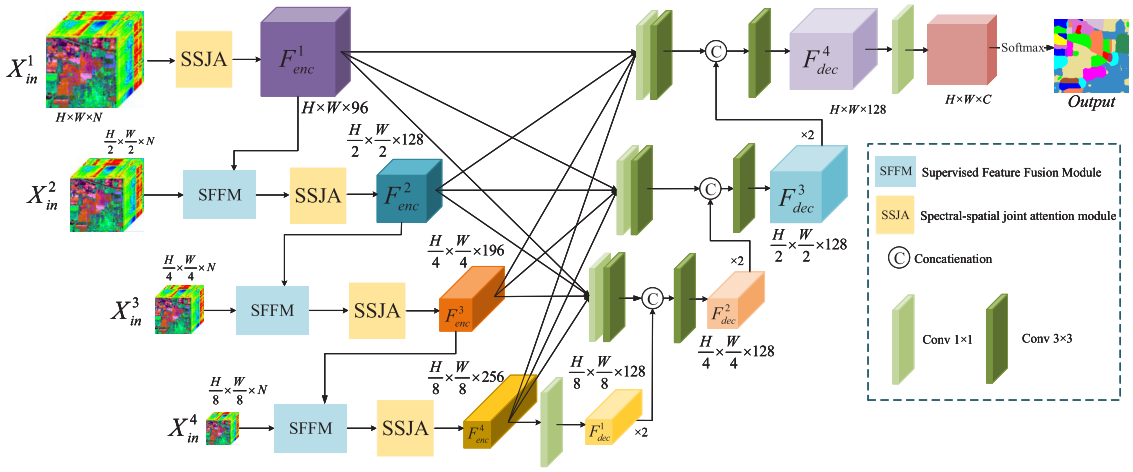

Fig. 1. Flowchart of the proposed method.

from the previous stage encoder layer and a downsampled HSI.

We denote the HSI as $X_{\text{in}} \in \mathbb{R}^{H\times W\times N}$, where $H \times W$ is the dimension of HSI and N is the number of bands, while $X^i_{\text{in}} \in \mathbb{R}^{(H/2^{i-1})\times(M/2^{i-1})\times N}$ is the input HSI pyramid at the ith level of encoder and obtained by the bicubic interpolation operation. Moreover, F^i_{enc} and F^i_{dec} are semantic features from the encoder and the decoder at the ith level. The initial HSI input X^1_{in} is delivered to the first encoder layer consisting of a convolutional layer and SSAJ to extract feature F^1_{enc}. We stacked SFFM and SSJA to build the rest of the encoder layers. The SFFM is plugged between every two encoder layers. Then, we have

$$F^1_{\text{enc}} = H_{\text{SSJA}}\left(H^{3\times3}_{\text{conv}}\left(X^1_{\text{in}}\right)\right) \tag{1}$$

$$F^i_{\text{enc}} = H_{\text{SSJA}}\left(H_{\text{SFFM}}\left(X^i_{\text{in}}, F^{i-1}_{\text{enc}}\right)\right) \tag{2}$$

where $i \in \{2, 3, 4\}$ and $H^{3\times3}_{\text{conv}}$ represents the 3×3 convolutional layer. $H_{\text{SSJA}}(\cdot)$ denotes our proposed the SSJA module, and $H_{\text{SFFM}}(\cdot)$ denotes our proposed SFFM.

2) Multiple-to-One Connection: In most encoder–decoder HSI classification networks [29], [30], only the features from the coarser scale subnetwork are flowing to the finer-scale subnetworks, making information flow inflexible.

In our CLSJE, we encourage the coarser-to-finer and finer-to-coarser information flow. Unlike the intermediate supervision in the conventional coarse-to-fine networks, our CLSJE applies the intermediate supervision to each decoder layer. Specifically, these multiscale feature maps from different encoder layers and the decoded feature maps from the coarser scale are aggregated to the corresponding decoder layer. The outputs of all encoder layers are concatenated. Then, the multiscale features are processed using two stacks of 1×1 and 3×3 convolutional layers. Note that the interpolating operation is applied such that the features from different scales can be concatenated. Then, multiscale features and the upsampled decoded feature are concatenated and transported to 3×3 convolutional layers. The decoded feature maps in each decoder layer can be expressed by the following formula:

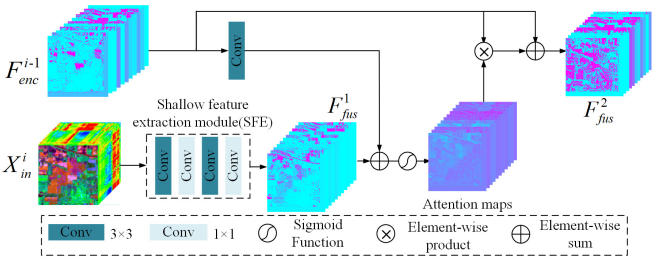

Fig. 2. Schematic of SFFM.

$$F^1_{\text{dec}} = H^{1\times1}_{\text{conv}}\left(F^4_{\text{enc}}\right) \tag{3}$$

$$F^i_{\text{dec}} = H^{3\times3}_{\text{conv}}\left(\text{Cat}\left(H_{\text{conv2}}\left(\text{Cat}\left(F^1_{\text{enc}}, F^2_{\text{enc}}, F^3_{\text{enc}}, F^4_{\text{enc}}\right)\right)\right.\right.$$
$$\left.\left.H_{\uparrow}\left(F^{i-1}_{\text{dec}}\right)\right)\right) \tag{4}$$

where $\text{Cat}(\cdot)$ denotes the concatenation operation, H_{conv2} denotes two stacks of 1×1 and 3×3 convolutional layers, and $H^{1\times1}_{\text{conv}}$ represents the 1×1 convolutional layer.

B. Supervised Feature Fusion Module

It is insufficient that different-level features with different receptive fields are directly concatenated or added. Instead, we introduce SFFM between the two encoder layers, facilitating the expected feature stream generation and achieving significant performance gain. The schematic of SFFM is demonstrated in Fig. 2, and its contributions are twofold. First, it provides fine-level supervisory signals used for coarse-scale HSI feature extraction. Second, under the supervision of global information, we introduce attention maps to activate the discriminated features and suppress the less informative features. The fused features only include useful ones that are allowed to propagate to the encoder layer.

SFFM takes the downsampled HSI and the encoded features $F^i_{\text{enc}} \in \mathbb{R}^{(H/2^{i-1})\times(M/2^{i-1})\times C}$ from the earlier encoder layer as inputs, where C is the number of channels. SFFM first extracts the shallow feature of the downsampled HSI X^i_{in} via a shallow

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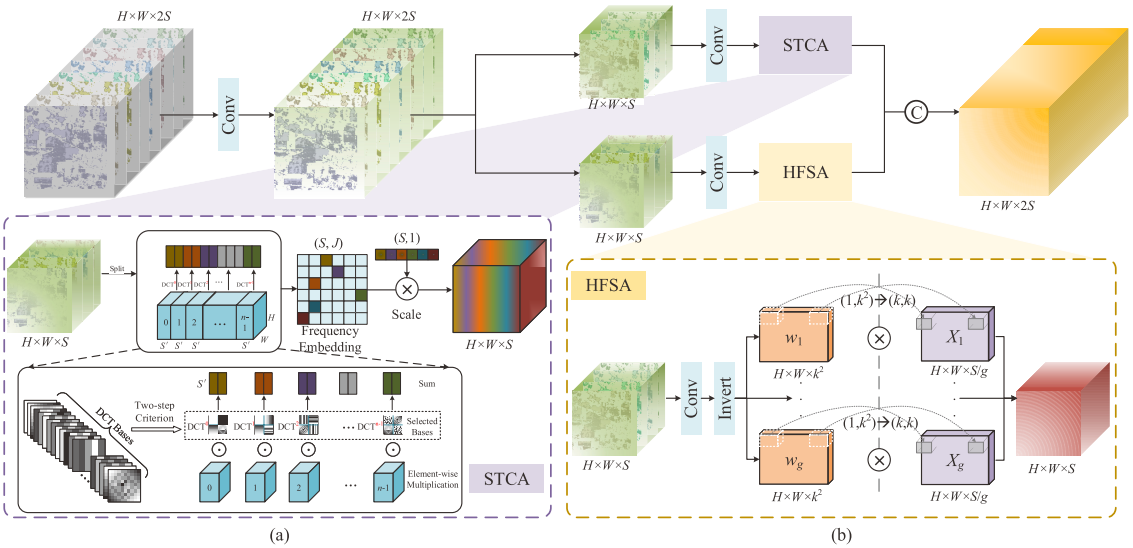

Fig. 3. Details of the SSJA. (a) Flowchart of the STCA. (b) Detail architecture of the HFSA.

feature extraction (SFE) module, which contains two 1×1 convolutions attached to 3×3 convolutions. Next, the shallow feature is added to the encoded features to obtain the fused features $F^1_{\text{fus}} \in \mathbb{R}^{(H/2^{i-1})\times(M/2^{i-1})\times C}$ of the first stage. The fused features aggregate both local and global features corresponding to the downsampledd HSI X^i_{in} and the encoded features F^{i-1}_{enc}. Per-pixel attention maps $M \in \mathbb{R}^{(H/2^{i-1})\times(M/2^{i-1})\times C}$ are generated from fused features F^1_{fus} using the sigmoid activation. The useful information is activated and propagated to supervise the next-stage feature fusion. These maps are then used to recalibrate the encoded features, resulting in attention-guided fused features. Finally, the fused feature F^2_{fus} produced by SFFM is transported to SSJA.

C. Spectral–Spatial Joint Attention Module

Current CNN models exhibit a significant flaw: these models tend to preferentially learn low-frequency components. However, the discriminated features are often from the HF signal responses. Compared to RGB images, HSIs contain redundant spectral information increasing the difficulty of learning discriminated features. Therefore, HF signal responses are more critical for HSI classification. SSJA is the essential part of each encoder layer, which can extract precious spectral–spatial features bypassing abundant less informative features, as shown in Fig. 3. The SSJA contains HFSA and STCA. SSJA explicitly wires spectral information and spatial information as two separate processing branches (i.e., spectral and spatial streams) and then processes information in parallel. HFSA and STCA are used to reweight the spatial features and spectral features, respectively, and focus on the HF components simultaneously.

1) High-Frequency Spatial Attention: The frequency of a signal varies randomly in spatial locations [56]. Therefore, only using convolution operation is unable to capture all the HF signals. HFSA captures the HF signals by actively applying different high-pass kernels across spatial locations. HFSA learns to generate different high-pass kernels corresponding to the spatial locations. The detailed network architecture is shown in Fig. 3(b). We split the channels into groups, which reduces channel feature redundancy and even computational overheads. Given an input feature map $X \in \mathbb{R}^{H\times W\times S}$, we split the S channels into g groups. The input feature X is then processed by a 3×3 convolution and batch normalization. We first applied the softmax function to get the low-pass filters [57]. Then, we obtain g high-pass kernels $\omega \in \mathbb{R}^{H\times W\times k^2}$ for each spatial location by inverting the low-pass kernel. Finally, we apply these high-pass kernels to X all groups to generate the output Y.

2) Spectral-Transform Channel Attention: Most attention modules (i.e., GJAM [30], spectral attention [29], and so on) applied to HSI classification calculate the average value of each channel using global average pooling (GAP). However, GAP only uses the lowest frequency information of the input features [58]. Thus, these HSI classification networks using the GAP component likely discard other potentially useful HF features. We claim that these HF components carry the most discriminative features. Considering this flaw, we propose STCA that transforms spectral components into frequency components to recalibrate HF features at different spectrums.

Following the design approach of [58], the feature is transformed to the frequency domain using DCT. We transform the feature maps to a frequency embedding using the DCT. Given an input $x \in \mathbb{R}^{H\times W}$, H is the height of x, and W is the width of x. The 2-D DCT frequency spectrum $f^{2D} \in \mathbb{R}^{H\times W}$ can be written as

$$f^{2D}_{h,w} = \sum_{i=0}^{H-1}\sum_{j=0}^{W-1} x_{i,j} \underbrace{\cos\left(\frac{\pi h}{H}\left(i+\frac{1}{2}\right)\right)\cos\left(\frac{\pi w}{W}\left(j+\frac{1}{2}\right)\right)}_{\text{DCT weight}}$$
$$\text{s.t. } h \in \{0,1,\ldots,H-1\}, w \in \{0,1,\ldots,W-1\}. \quad (5)$$

Note that, in (5), the normalizing constants are omitted for simplicity, which would not affect the results in this work.

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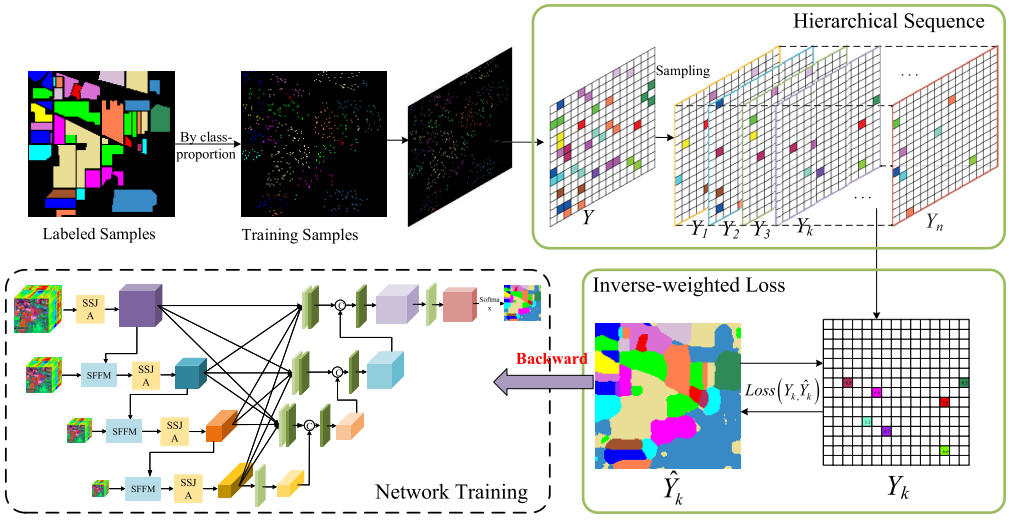

Fig. 4. Procedure of the proposed CSS with inverse-weighted loss.

As discussed in FCANet [58], $f_{h,w}^{2D}$ produced by 2-D-DCT can be viewed as a weighted sum of inputs.

The specifics of STCA are described in Fig. 3(a). Supposed an input features $X \in \mathbb{R}^{H\times W\times S}$, we split X into many parts along the channel dimension $X^z \in \mathbb{R}^{H\times W\times D}$, $z \in \{0, 1, \ldots, n-1\}$, $D = (S/n)$. We want to use J frequency components, so we use (6) to calculate the matrix of DCT weights $W \in \mathbb{R}^{H\times W\times D\times J}$

$$W_{r,i,j} = \cos\left(\frac{\pi u(2i+1)}{2H}\right)\cos\left(\frac{\pi v(2j+1)}{2H}\right) \tag{6}$$

where $r \in \{0, 1, 2, \ldots, J\}$ and corresponds to a specific component (u, v). Then, the elementwise multiplication and a spatial sum are performed to produce our DCT coefficients $U \in \mathbb{R}^{D\times J}$

$$U^z = \sum_{i=0}^{H-1}\sum_{j=0}^{W-1} X^z_{:,:,i,j} \odot A^T_{:,:,i,j}. \tag{7}$$

To reduce the dimension of DCT coefficients, in each channel, we select the maximum frequency response

$$U_C = \max\left(U^0, U^1, U^2, \ldots, U^{n-2}, U^{n-1}\right). \tag{8}$$

We then apply the fully connected layer followed by a standard sigmoid $\sigma(x)$ to calculate the attention map

$$\text{Att}_{\text{STCA}} = \sigma(F(U_C)). \tag{9}$$

Finally, the attention map Att_{STCA} is used to reweight the input features.

D. Class-Proportion Sampling Strategy

HSI classification suffers from imbalanced training samples, which needs to be processed through appropriate sampling and loss strategy. The CSS is adopted to speed model convergence and balance sample weights. The overall flowchart of the CSS is shown in Fig. 4. The proposed CSS consists of the hierarchical sequence and the inverse-weighted cross-entropy loss.

Considering the strong sparsity of HSI training samples, we selected the training samples of each class by proportion and input them into the hierarchical sequence in random order. The behavior of the minibatch sampler is simulated via class-proportion sampling to obtain stable yet diverse gradients. Specifically, we first merge all training pixels of various classes rather than the training patches of the local areas by proportion in a list indexed as class labels. Second, to keep the combination stochasticity, the training samples of each class in the list are randomly assigned to a hierarchical sequence. $Y_k(k = 1, 2, \ldots, n)$ denotes the kth layer of the hierarchical sequence, which contains partial samples of all classes. The samples of all layers compose the entire training samples. The hyperparameter α, which denotes the minibatch per class, is introduced to maintain the training stability. Within a range, the smaller the value of α, the greater the number of the obtained stochastic gradient orientations, and the faster the model convergence. Notably, the class imbalance of training samples is one of the main obstacles affecting the HSI classification accuracy.

Limited by the difficulty in visually identifying many land covers, the number of labeled HSI samples per class varied. The inverse-weighted cross-entropy loss is proposed to balance the probability of each class by adding the inverse-weighted factor to the standard cross-entropy criterion. In more detail, we increase the relative loss weight for easily misclassified classes and reduce the relative loss weight for well-classified classes. The loss function for the ith class is defined as

$$\text{loss}(i) = \frac{\sum_{i=1}^{c}\text{num}_i}{c^*\text{num}_i} \tag{10}$$

where c denotes the total class number of the input HSI dataset and num_i denotes the number of training samples for class i. The proposed sampling strategy makes it possible to speed up

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model convergence and improve the classification accuracy of the difficult-classified samples.

IV. Experimental Evaluation

In this section, we implement experiments on four benchmark datasets to illustrate the effectiveness of the proposed CLSJE. The nine-class ROSIS Pavia University (PU) dataset, the 16-class IP dataset, the 15-class Compact Airborne Spectrographic Imager University of Houston dataset, and the 14-class Botswana dataset are utilized to validate the performance of our proposed classification method in the case of inadequate and imbalance samples. The classifiers used in the comparison are nine state-of-the-art methods (CDCNN [59], FDSSC [60], SSFCN [49], SSRN [40], SSTN [61], DBMA [62], DBDA [63], FPGA [29], and SSDGL [30]). All of the experiments presented in this section are performed on RTX3090.

A. Experimental Settings

1) Optimized Parameters: The proposed CLSJE was trained 600 iterations of training processes for the PU dataset, the IP dataset, and the Botswana dataset, while 1000 iterations of training processes for the Houston dataset and 2018 IEEE GRSS DFC Dataset. For PU dataset, IP dataset, Botswana dataset, Houston dataset, and 2018 IEEE GRSS DFC Dataset, the proposed CLSJE using gradient descent with momentum 0.9 and the weight decay rate 0.001, the initial learning rate was set to 0.005 and multiplied by $(1 - \text{iteration}/\text{max_iteration})^{\text{power}}$, power $= 0.8$.

2) Metrics: The classification performances of the proposed method and the state-of-the-art methods are evaluated by the overall accuracy (OA), average accuracy (AA), kappa coefficient (Kappa), and the accuracy per class. All the quantitative results are presented in the form of mean value $\pm$ standard deviation by choosing random samples and calculating the result of ten times experiments.

B. Experiment 1: Pavia University Dataset

The PU dataset containing 610 $\times$ 340 pixels with a geometric resolution of 1.3 m was acquired by the ROSIS sensor during a flight campaign over Pavia, Northern Italy. The number of spectral bands is 103 with the removal of 12 bands covering the region of noise and water absorption. This dataset contains nine classes representing the different land-cover types. The CSS randomly selects 1% of labeled samples as training data, and the others become test data, which is listed in Table I. Notably, the minimum number of training samples per class is set to 5. The false-color composition picture and the corresponding ground-truth map are shown in Fig. 5 as a visual reference.

For quantitative evaluation on the PU dataset, the accuracy per class, OA, AA, and Kappa are, respectively, listed in Table II under the same training samples configuration, in which the best results are highlighted in bold. It can be seen that our method keeps the highest classification accuracy among all the methods. SSDGL uses the global convolutional long short-term memory to extract the interdependency of

TABLE I
Numbers of Training and Test Samples for the PU Dataset

Class number	Class name	Training	Test	Total
1	Asphalt	67	6564	6631
2	Meadows	187	18462	18649
3	Gravel	21	2078	2099
4	Trees	31	3033	3064
5	Metal Sheets	14	1331	1345
6	Bare Soil	51	4978	5029
7	Bitumen	14	1316	1330
8	Bricks	37	3645	3682
9	Shadows	10	937	947
Total		432	42344	42776

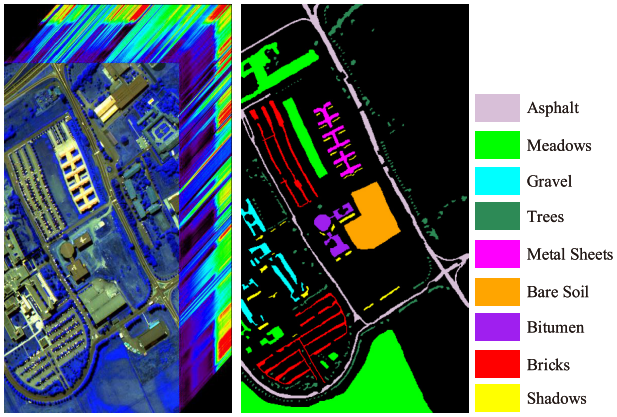

Fig. 5. False-color image and ground-truth map for the PU dataset.

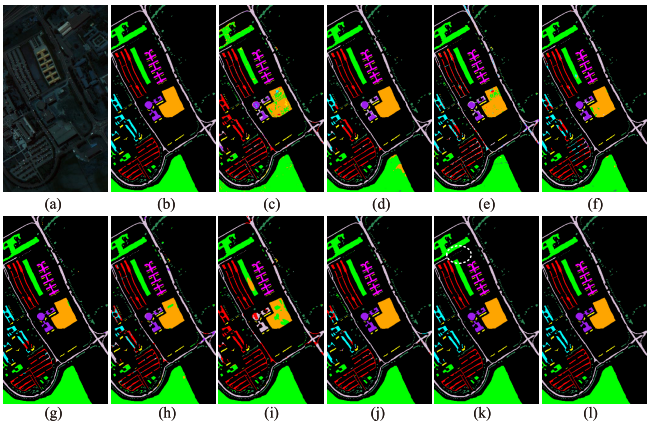

Fig. 6. Visualization of the classification maps for the PU dataset. (a) False-color image. (b) Ground truth. (c) CDCNN. (d) FDSSC. (e) SSFCN. (f) SSRN. (g) SSTN. (h) DBMA. (i) DBDA. (j) FPGA. (k) SSDGL. (l) CLSJE.

channels, which obtains the classification accuracy second only to our method. In contrast, our CLA architecture and the proposed SSJA mechanism are more conducive to guaranteeing the complementarity and abundance of extracted features.

In order to intuitively show the performance of our method, we further exhibit the visual results of the classification map. As can be observed from Fig. 6, the classification map of our method generates more distinct class blocks and fewer misclassified pixels than that of other methods. CDCNN misclassifies gravel into bricks, leading to the poor performance of classification on this dataset. The classification map of SSFCN

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TABLE II
CLASSIFICATION RESULTS ON THE PU DATASET WITH 1% LABELED SAMPLES (BEST RESULTS ARE SHOWN IN BOLD)

Class number	CDCNN	FDSSC	SSFCN	SSRN	SSTN	DBMA	DBDA	FPGA	SSDGL	CLSJE
1	85.74±5.50	96.88±1.50	87.84±3.08	97.41±2.05	86.63±5.95	96.26±3.45	89.03±2.21	97.83±0.21	**100.00±0.00**	**100.00±0.00**
2	94.45±3.76	97.57±2.16	98.05±1.62	99.10±0.20	98.77±3.74	98.31±2.11	98.32±1.34	99.95±0.14	**100.00±0.00**	**100.00±0.00**
3	32.59±12.47	89.97±4.90	74.78±14.74	88.61±5.40	86.16±8.81	89.16±9.94	98.70±1.65	91.28±0.55	**100.00±0.00**	**100.00±0.00**
4	97.46±4.29	99.21±1.47	88.72±7.31	99.81±0.45	86.70±2.47	97.53±2.19	98.42±0.54	95.85±0.35	99.67±0.21	**99.84±0.07**
5	99.10±1.99	99.55±0.32	93.91±4.43	**100.00±0.00**	**100.00±0.00**	99.72±1.25	99.78±0.49	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
6	80.88±11.34	97.97±1.63	89.17±7.82	95.51±2.80	98.48±1.97	97.98±1.09	98.57±0.61	99.76±0.21	**100.00±0.00**	**100.00±0.00**
7	88.83±6.82	**100.00±0.00**	69.77±5.33	92.16±3.62	86.58±8.90	85.51±8.40	95.84±2.59	99.73±0.10	**100.00±0.00**	**100.00±0.00**
8	66.19±11.38	70.97±3.27	78.93±5.04	89.03±4.17	96.51±3.21	80.70±6.90	89.47±3.09	98.05±0.02	99.92±0.01	**100.00±0.00**
9	96.01±1.08	**100.00±0.00**	75.13±1.53	99.96±0.15	65.24±9.15	91.12±8.90	99.89±0.21	97.86±0.05	**100.00±0.00**	**100.00±0.00**
OA (%)	87.70±2.59	94.43±1.89	90.45±1.33	96.55±0.73	93.73±5.34	95.13±1.24	96.00±0.65	98.68±0.02	99.97±0.01	**99.99±0.01**
AA (%)	82.36±7.02	94.68±1.11	84.03±1.70	95.73±1.03	89.45±7.58	93.48±1.88	96.45±0.77	97.82±0.02	99.95±0.01	**99.99±0.00**
Kappa	0.8359±0.0368	0.9257±0.0253	0.8729±0.0181	0.9543±0.0096	0.9210±0.0737	0.9353±0.0169	0.9467±0.0087	0.9825±0.01	0.9996±0.0011	**0.9998±0.0001**

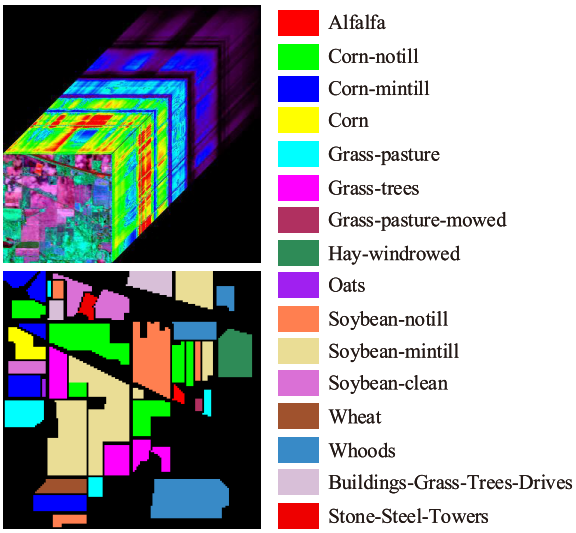

Fig. 7. False-color image and ground-truth map for the IP dataset.

TABLE III
NUMBERS OF TRAINING AND TEST SAMPLES FOR THE IP DATASET

Class number	Class name	Training	Test	Total
1	Alfalfa	5	41	46
2	Corn-notill	72	1356	1428
3	Corn-mintill	42	788	830
4	Corn	12	225	237
5	Grass-pasture	25	458	483
6	Grass-trees	37	693	730
7	Grass-pasture-mowed	5	23	28
8	Hay-windrowed	24	454	478
9	Oats	5	15	20
10	Soybean-notill	49	923	972
11	Soybean-mintill	123	2332	2455
12	Soybean-clean	30	563	593
13	Wheat	11	194	205
14	Whoods	64	1201	1265
15	Buildings-Grass -Trees-Drives	20	366	386
16	Stone-Steel-Towers	5	88	93
	Total	529	9720	10249

has relatively clear profiles but still possesses severe noise. The reason is that the pixel-to-pixel FCN proposed by SSFCN requires adequate training samples. In contrast, the network cannot reach the best classification effect by training 1% of the labeled samples. Due to the original 3-D pixel data input limitation, DBMA and DBDA produce a little noise in the classification map. FPGA and SSDGL can generate better classification results through the assistance of the patch-free encoder–decoder network. However, the one-to-one encoder–decoder connection would lose some significant spectral–spatial features in the network modules, which negatively influences HSI classification. Compared with these methods, our method can classify HSI accurately with an OA of 99.99%.

C. Experiment 2: Indian Pines Dataset

This dataset was gathered by AVIRIS sensor over the IP test site in Northwestern Indiana and consisted of 145 × 145 pixels and 220 bands. Removing 20 bands covering the region of water absorption, 200 bands are retained for the classification task. The available ground truth is designed into 16 classes and is not all mutually exclusive. In the experiment, 5% of labeled samples are trained for the network, and Fig. 7 illustrates the false-color image and corresponding ground-truth map. Table III lists the numbers of training and test samples, in which the classes of Alfalfa, Grass-pasture-mowed, and Oats have only 46, 28, and 20 samples, respectively. The highly imbalanced distribution of the labeled samples is in the IP datasets' various classes, which leads to the training difficulty.

The visual comparison of classification results with different methods is presented in Fig. 8. It can be seen that FPGA, SSDGL, and our method have better visual results than other methods. The classification results by CDCNN and SSFCN are barely satisfactory. This is because the significant intraclass variations and the slight interclass dissimilarity existed in this dataset, exacerbating the difficulty of HSI classification. SSTN and FPGA reveal the passive effect of HSI classification in classes with similar spectral features, such as corn and soybean. Besides, SSDGL still generates classification results with small amounts of noise in the region of soybean and grass. Comparatively, our method has superior performance and better generalization ability.

With regard to the quantitative indicators for HSI classification shown in Table IV, our method outperforms the other methods by approximately 0.36%–27.35%, 0.2%–27.76%, and 0.004–0.3125 in terms of OA, AA, and Kappa, respectively. Through the proposed CLSJE architecture, our method can

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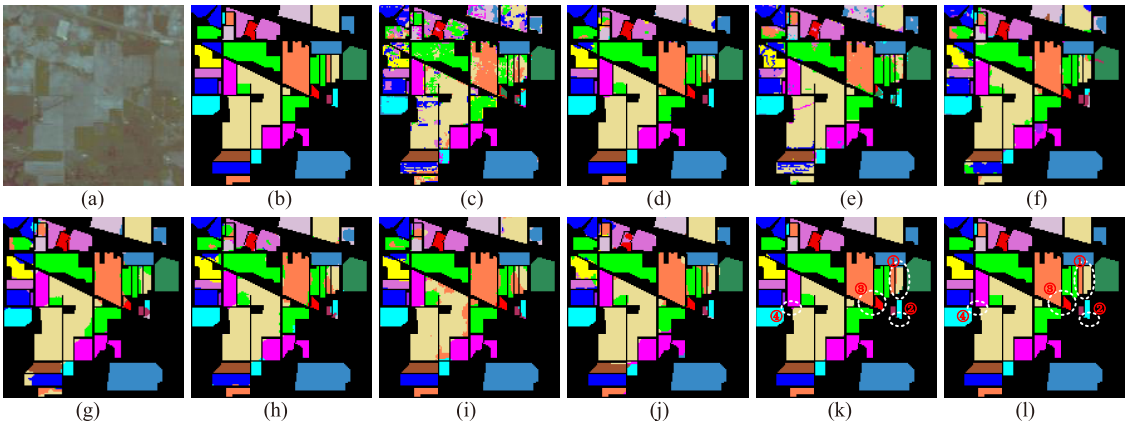

Fig. 8. Visualization of the classification maps for the IP dataset. The white dashed lines outline part of the subtle classification distinctions. (a) False-color image. (b) Ground truth. (c) CDCNN. (d) FDSSC. (e) SSFCN. (f) SSRN. (g) SSTN. (h) DBMA. (i) DBDA. (j) FPGA. (k) SSDGL. (l) CLSJE.

TABLE IV
CLASSIFICATION RESULTS ON THE IP DATASET WITH 5% LABELED SAMPLES (BEST RESULTS ARE SHOWN IN BOLD)

Class number	CDCNN	FDSSC	SSFCN	SSRN	SSTN	DBMA	DBDA	FPGA	SSDGL	CLSJE
1	29.80±5.17	85.42±7.88	73.17±12.13	75.57±10.54	92.36±4.28	90.37±7.43	**100.00±0.00**	97.22±0.01	**100.00±0.00**	**100.00±0.00**
2	67.42±13.31	97.20±3.38	83.42±4.04	90.65±2.73	88.96±4.11	92.72±3.92	88.49±8.61	93.07±0.04	99.63±0.01	**100.00±0.00**
3	57.79±12.51	94.45±3.54	89.16±2.01	97.01±3.46	71.79±5.73	95.63±3.48	97.12±1.63	89.46±0.07	99.24±0.02	**100.00±0.00**
4	64.16±7.76	**100.00±0.00**	75.12±8.93	93.36±3.00	54.96±14.69	89.35±5.46	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
5	94.46±5.21	**100.00±0.00**	88.02±5.95	98.56±1.84	72.32±4.27	96.92±2.86	**100.00±0.00**	95.63±0.05	99.56±0.00	**100.00±0.00**
6	91.23±4.73	**100.00±0.00**	93.00±1.59	98.94±1.31	93.55±1.08	99.18±0.43	97.18±0.95	97.56±0.00	**100.00±0.00**	**100.00±0.00**
7	63.26±13.96	73.53±5.31	40.00±19.02	84.21±5.02	79.26±10.12	79.57±14.25	92.59±4.42	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
8	88.88±2.72	99.78±1.44	96.28±2.77	98.36±1.48	99.90±2.79	99.11±0.46	99.78±0.31	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
9	63.02±11.87	**100.00±0.00**	67.78±15.61	97.61±4.70	50.53±20.69	97.91±2.31	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
10	65.63±9.54	89.25±3.66	79.29±4.24	81.03±2.87	90.85±4.16	92.08±4.81	89.87±8.46	96.64±0.03	99.68±0.02	**100.00±0.00**
11	73.28±7.68	93.97±0.82	90.77±3.16	93.02±2.19	97.08±3.85	95.15±4.60	99.33±0.56	96.74±0.00	98.36±0.04	**100.00±0.00**
12	46.85±10.6	95.41±3.15	88.37±2.64	95.72±3.95	76.85±2.81	90.71±4.70	98.50±1.12	91.65±0.12	99.11±0.05	**99.82±0.01**
13	90.32±7.19	**100.00±0.00**	94.57±2.02	99.81±0.32	93.33±1.11	99.81±0.00	96.02±3.27	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
14	90.19±4.10	93.14±1.69	96.31±1.81	95.79±1.59	95.12±1.72	97.11±1.62	93.22±4.55	99.91±0.00	**100.00±0.00**	**100.00±0.00**
15	76.53±7.40	90.61±2.94	66.57±5.33	92.25±3.11	87.02±2.38	88.13±8.38	96.99±2.21	99.72±0.02	**100.00±0.00**	**100.00±0.00**
16	92.93±6.44	96.55±1.95	84.34±8.82	96.57±2.29	84.95±2.17	97.05±4.12	94.38±4.84	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
OA (%)	72.64±6.44	94.87±0.86	87.76±3.97	92.21±0.84	91.20±1.26	94.43±1.60	95.38±1.43	96.18±0.02	99.63±0.02	**99.99±0.01**
AA (%)	72.23±5.46	94.33±1.35	72.83±3.51	93.03±1.35	83.05±2.80	93.81±1.99	96.47±1.50	97.33±0.01	99.79±0.00	**99.99±0.00**
Kappa	0.6873±0.0508	0.9414±0.0098	0.8604±0.0259	0.9115±0.0257	0.8985±0.0145	0.9365±0.0184	0.9474±0.0148	0.9564±0.0005	0.9958±0.0003	**0.9998±0.0001**

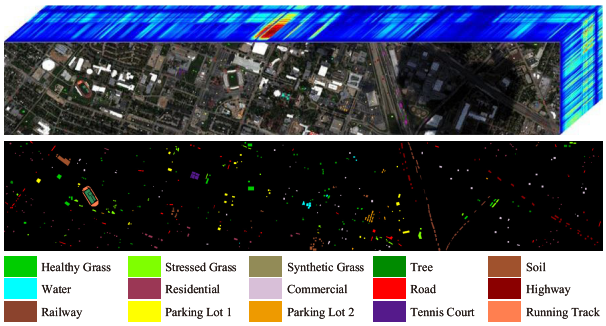

Fig. 9. False-color image and ground-truth map for the HU dataset.

obtain multiscale discriminative feature representations with complementary properties to reduce the intraclass distance and increase the interclass distance. Consequently, our method yields the most satisfactory description among all mentioned methods on this highly imbalanced dataset.

TABLE V
NUMBERS OF TRAINING AND TEST SAMPLES FOR THE HU DATASET

Class number	Class name	Training	Test	Total
1	Healthy Grass	10	1241	1251
2	Stressed Grass	10	1244	1254
3	Synthetic Grass	10	687	697
4	Tree	10	1234	1244
5	Soil	10	1242	1252
6	Water	10	315	325
7	Residential	10	1258	1268
8	Commercial	10	1234	1244
9	Road	10	1242	1252
10	Highway	10	1217	1227
11	Railway	10	1225	1235
12	Parking Lot 1	10	1224	1234
13	Parking Lot 2	10	459	469
14	Tennis Court	10	418	428
15	Running Track	10	650	660
	Total	150	14861	15011

D. Experiment 3: Houston University Dataset

The Houston University (HU) dataset was published in the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) data fusion contest, which is a relatively tricky benchmark dataset. The dataset contains 15 land-cover types with 349 × 1905 pixels and 144 spectral bands ranging

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TABLE VI
CLASSIFICATION RESULTS ON THE HU DATASET WITH TEN LABELED SAMPLES PER CLASS (BEST RESULTS ARE SHOWN IN BOLD)

Class number	CDCNN	FDSSC	SSFCN	SSRN	SSTN	DBMA	DBDA	FPGA	SSDGL	CLSJE
1	76.78±17.69	88.57±6.31	86.51±8.52	74.37±9.49	98.62±1.91	85.32±8.39	86.11±5.85	97.90±0.93	99.84±0.16	**100.00±0.00**
2	81.56±15.82	94.61±5.35	85.98±9.84	78.41±8.71	99.27±0.94	90.87±7.36	95.62±3.51	87.54±5.25	98.71±0.20	**100.00±0.00**
3	56.94±24.37	99.80±0.60	96.81±2.17	73.26±10.65	99.85±0.13	99.15±0.25	99.91±0.21	**100.00±0.00**	99.85±0.00	**100.00±0.00**
4	86.55±17.24	99.38±0.93	78.07±9.87	78.53±8.07	93.79±3.46	93.79±8.12	98.51±2.06	98.22±1.31	97.16±0.24	**100.00±0.00**
5	95.04±2.50	96.33±1.78	92.76±5.81	92.56±3.76	95.09±3.23	93.63±3.61	96.42±1.78	99.92±0.87	**100.00±0.00**	**100.00±0.00**
6	84.83±16.15	99.84±0.47	26.48±6.60	76.94±10.26	83.61±7.04	99.61±0.72	99.58±0.66	93.97±2.65	98.10±0.45	**100.00±0.00**
7	66.53±12.92	85.02±4.55	60.08±11.45	54.27±14.46	66.91±14.60	73.57±18.46	84.91±10.42	84.34±0.32	98.97±1.04	**99.68±0.21**
8	78.05±11.46	94.10±6.57	63.77±18.11	92.89±4.25	58.74±18.09	82.16±15.97	91.28±6.23	66.12±3.00	66.29±1.08	**100.00±0.00**
9	64.91±11.55	78.62±7.13	62.47±22.1	68.58±15.79	90.75±5.62	79.06±5.89	83.09±7.45	91.06±4.82	92.03±0.04	**99.76±0.010**
10	46.19±16.82	71.53±7.12	74.55±10.29	65.11±17.14	90.72±5.74	71.47±7.84	74.54±16.07	96.63±2.79	99.92±0.00	**100.00±0.00**
11	52.40±12.07	86.95±8.87	83.55±15.87	82.07±8.94	76.05±9.93	79.81±14.13	85.70±8.71	68.82±7.51	93.63±0.45	**100.00±0.00**
12	57.18±11.53	83.27±12.09	81.23±10.01	74.63±11.29	57.30±13.82	79.67±7.34	79.66±7.24	90.60±2.09	99.67±0.75	**100.00±0.00**
13	80.12±11.29	84.25±13.71	15.73±6.15	73.77±9.06	95.10±3.03	86.93±7.48	87.59±10.11	96.29±1.38	95.86±1.21	**100.00±0.00**
14	76.14±11.91	97.39±3.12	83.22±9.04	74.83±14.93	**100.00±0.00**	98.80±1.95	98.03±3.00	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
15	82.96±10.22	96.85±2.22	40.43±5.76	89.12±2.38	**100.00±0.00**	95.38±1.89	94.51±2.37	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
OA (%)	68.79±4.50	87.98±1.75	73.37±3.95	74.51±1.76	85.15±1.27	83.65±3.89	88.20±1.84	89.89±0.75	95.36±0.64	**99.95±0.03**
AA (%)	72.41±4.19	90.43±1.24	68.78±5.42	77.18±1.32	87.05±1.17	87.28±2.46	90.36±1.46	91.43±0.64	96.00±0.25	**99.96±0.02**
Kappa	0.6626±0.0489	0.8700±0.0189	0.7114±0.0431	0.7235±0.0190	0.8395±0.0137	0.8231±0.0222	0.8723±0.0199	0.8908±0.0037	0.9499±0.0014	**0.9995±0.0002**

from 380 to 1050 nm. Ten labeled samples per class are selected to train the proposed network. Fig. 9 shows the false-color image and the corresponding ground-truth map. The number of training and test samples is listed in Table V. The insufficiency of training samples in this dataset increases the difficulty of HSI classification and demonstrates the strong performance of the proposed method in this case.

It can be seen from Table VI that the classification accuracies of these methods reveal various classification quality levels. CDCNN as a typical HSI classification method shows an unsatisfactory classification quality. FDSSC, SSTN, DBMA, and DBDA methods yield a comparable classification effect but still generate low classification accuracy in some classes. The global learning framework that FPGA, SSDGL, and our method adopt achieves better classification accuracy on OA, AA, and Kappa than the patch-based methods, which has a remarkable breakthrough in the small sample classification. Notably, our method possesses the saturation of the classification accuracy per class except for the classes of residential and road. Our method tackles the loss of discriminative spectral information, specifically effective for scenes with small sample HSI classification.

The strong performance of our method is shown in Fig. 10 from the visual perspective. DBMA and DBDA may generate oversmoothing phenomena or even misclassify the image when dealing with isolated targets with small sizes. In light of the white dashed lines in Fig. 10, FPGA and SSDGL still produce salt-and-pepper noise. The proposed method maintains good boundaries of narrow targets like railways by balancing the object continuity and the boundary information. The introduction of the SSJA mechanism provides more discriminative spectral features, which assists in further improving the performance of our method. Overall, the proposed method outperforms other methods in terms of performance for small sample classification.

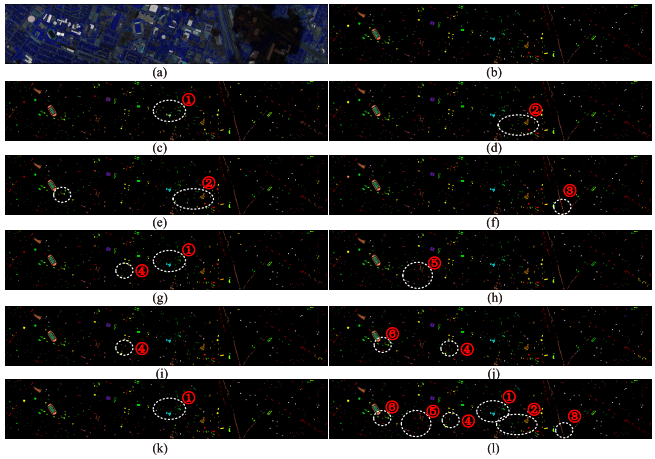

Fig. 10. Visualization of the classification maps for the HU dataset. The white dashed lines outline part of the subtle classification distinctions. (a) False-color image. (b) Ground truth. (c) CDCNN. (d) FDSSC. (e) SSFCN. (f) SSRN. (g) SSTN. (h) DBMA. (i) DBDA. (j) FPGA. (k) SSDGL. (l) CLSJE.

E. Experiment 4: Botswana Dataset

The Botswana dataset was captured by the NASA EO-1 Hyperion sensor over the Okavango Delta, Botswana, in 2001–2004. The dataset is 1476 × 256 pixels at 30-m pixel resolution and 242 bands covering the 400–2500-nm portion of the spectrum. After uncalibrated and noisy bands were removed, the dataset contains 145 bands and 14 identified land-cover classes located in the distal portion of the Delta. The false-color image and the corresponding ground-truth map are shown in Fig. 11, and the numbers of training and test samples are listed in Table VII. Five samples per class are used for training, which increases the difficulty of HSI classification.

The typical differences can be seen in Fig. 12 and Table VIII that our method obtains the closest result to the ground truth with 99.65% OA, 99.70% AA, and 0.9962 Kappa. In contrast, other methods yield salt-and-pepper noise and misclassification pixels. CDCNN based on 2-D-CNN achieves the worst accuracy with 80.90% OA due to the limited training samples and the weak network structure. Benefiting from the dense connection, FDSSC enhances the performance of the network

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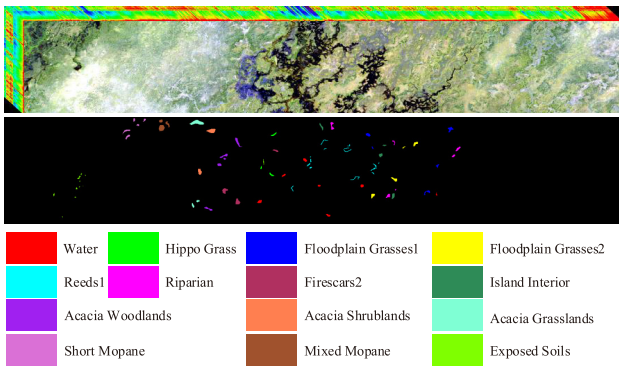

Fig. 11. False-color image and ground-truth map for the Botswana dataset.

TABLE VII

NUMBERS OF TRAINING AND TEST SAMPLES FOR THE BOTSWANA DATASET

Class number	Class name	Training	Test	Total
1	Water	5	265	270
2	Hippo Grass	5	96	101
3	Floodplain Grasses1	5	246	251
4	Floodplain Grasses2	5	210	215
5	Reeds1	5	264	269
6	Riparian	5	264	269
7	Firescars2	5	254	259
8	Island Interior	5	198	203
9	Acacia Woodlands	5	309	314
10	Acacia Shrublands	5	243	248
11	Acacia Grasslands	5	300	305
12	Short Mopane	5	176	181
13	Mixed Mopane	5	263	268
14	Exposed Soils	5	90	95
	Total	70	3178	3248

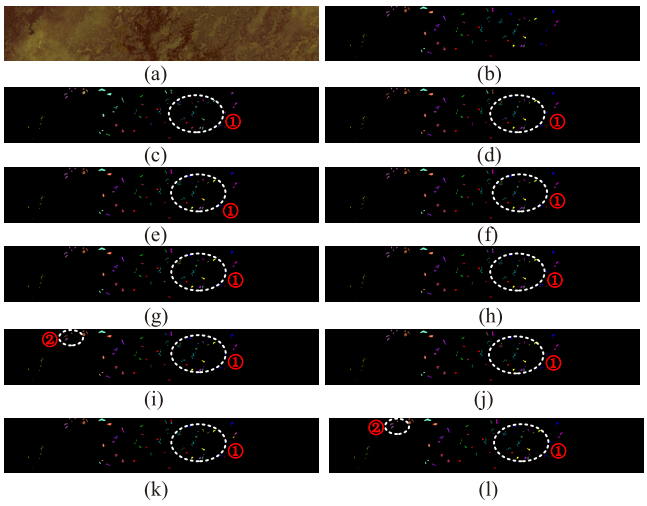

Fig. 12. Visualization of the classification maps for the Botswana dataset. The white dashed lines outline part of the subtle classification distinctions. (a) False-color image. (b) Ground truth. (c) CDCNN. (d) FDSSC. (e) SSFCN. (f) SSRN. (g) SSTN. (h) DBMA. (i) DBDA. (j) FPGA. (k) SSDGL. (l) CLSJE.

and obtains more than 10% improvement in OA compared to CDCNN. DBMA and DBDA extract the spectral–spatial feature in two independent branches, which accomplishes stable and reliable performance. SSFCN achieves high classification accuracy and reveals a comparable quality level with FPGA in this dataset. Based on the spectral–spatial residual blocks and the spectral–spatial transformer, SSRN and SSTN provide competitive results. In contrast, our method generates a more satisfactory description of the classification maps in the case of insufficient samples. Qualitative and quantitative results demonstrate that our method provides a substantial improvement on insufficient HSI classification. To gain insight into our method's effectiveness, we draw the confusion matrix on this dataset for quantitative evaluation. As shown in Fig. 13, the classification accuracy of each class is 100%, except for the Riparian class. It can be seen that the Riparian class tends to be wrongly classified as the Firescars2 class or the Island Interior class due to the similar shadow area of these classes. Besides, the small number of training samples makes it difficult to extract effective local spatial information from isolated pixels. Nevertheless, our method achieves excellent classification accuracy with higher than 95% accuracy per class.

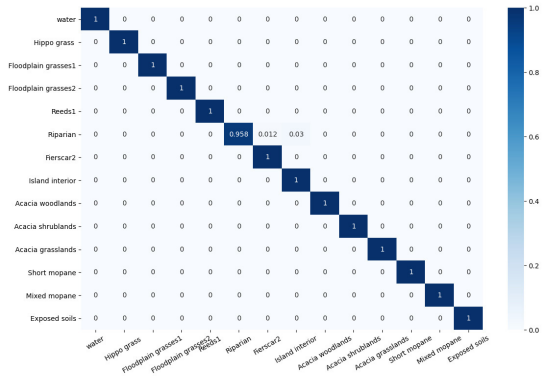

Fig. 13. Confusion matrix of the proposed method on the Botswana dataset.

F. Experiment 5: 2018 IEEE GRSS DFC Dataset

The 2018 IEEE GRSS DFC dataset was released by the Image Analysis and Data Fusion Technical Committee of the IEEE GRSS and the University of Houston for the 2018 data fusion contest, which holds more extensive training and test samples. This dataset has a larger spatial size of 601 × 2348 pixels with 48 spectral bands and contains 20 land-cover types. The false-color image and the corresponding ground-truth map are shown in Fig. 14. In the experiment, 5% of labeled samples are trained for our CLSJE. The detailed numbers of training and test samples in each class are listed in Table IX. Notably, the minimum number of training samples per class is set to 10.

This dataset has a large number of labeled samples, and the correlation between the training data and the test data is small, which can reflect the performance of classification methods well. As can be seen from Fig. 15, the misclassified pixels of CDCNN, SSFCN, SSRN, DBMA, and DBDA are obvious. FDSSC and SSTN yield a better classification effect. Compared with the above methods, FPGA and SSDGL achieve high classification accuracy and reveal comparable quality levels. In contrast, our method generates a more satisfactory

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TABLE VIII

CLASSIFICATION RESULTS ON THE BOTSWANA DATASET WITH FIVE LABELED SAMPLES PER CLASS (BEST RESULTS ARE SHOWN IN BOLD)

Class number	CDCNN	FDSSC	SSFCN	SSRN	SSTN	DBMA	DBDA	FPGA	SSDGL	CLSJE
1	94.60±2.89	97.41±2.06	99.59±0.87	99.23±1.43	95.00±2.74	97.77±2.31	95.64±2.45	96.04±2.12	**100.00±0.00**	**100.00±0.00**
2	68.64±11.94	98.95±12.51	88.89±5.78	**100.00±0.00**	**100.00±0.00**	88.89±6.91	98.99±1.86	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
3	81.11±6.89	**100.00±0.00**	95.11±4.6	**100.00±0.00**	**100.00±0.00**	100.0±0.00	100.0±0.00	98.37±0.84	98.37±2.87	**100.00±0.00**
4	65.45±15.54	93.03±6.73	93.26±3.87	**100.00±0.00**	**100.00±0.00**	92.51±4.52	91.30±3.88	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
5	89.10±4.74	80.74±7.1	87.19±6.91	77.22±10.88	86.10±6.70	93.51±3.84	95.58±2.99	66.29±5.62	72.35±6.47	**100.00±0.00**
6	69.28±13.18	84.93±5.2	90.91±6.36	99.23±0.86	99.61±1.07	68.94±9.64	82.23±7.40	99.24±0.37	26.14±1.54.	**95.75±0.21**
7	80.07±6.11	84.62±3.8	93.56±2.30	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	87.01±3.21	**100.00±0.00**	**100.00±0.00**
8	89.36±3.46	93.36±4.05	99.45±1.12	**100.00±0.00**	**100.00±0.00**	96.10±3.44	95.63±3.71	88.38±2.08	**100.00±0.00**	**100.00±0.00**
9	55.53±12.82	88.44±7.05	81.21±6.72	98.36±1.32	99.01±1.12	85.15±5.69	96.50±3.23	91.91±0.14	86.73±1.56	**100.00±0.00**
10	81.69±11.15	99.59±1.21	**100.00±0.00**	97.90±3.23	**100.00±0.00**	97.60±1.35	98.79±1.21	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
11	92.48±6.99	99.67±0.61	98.54±1.48	96.95±2.23	96.95±4.92	99.66±0.61	99.67±0.20	**100.00±0.00**	90.00±1.23	**100.00±0.00**
12	90.91±7.63	**100.00±0.00**	99.38±0.32	**100.00±0.00**	**100.00±0.00**	97.79±2.37	**100.00±0.00**	98.86±0.43	81.25±1.75	**100.00±0.00**
13	88.59±6.23	81.59±5.23	98.76±1.26	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
14	**100.00±0.00**	**100.00±0.00**	91.76±4.21	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	98.89±0.46	**100.00±0.00**
OA (%)	80.90±8.01	91.57±4.14	94.10±1.95	97.36±1.37	98.01±1.39	93.43±2.41	96.24±1.44	94.12±1.32	88.14±1.65	**99.65±0.18**
AA (%)	81.92±7.40	93.02±3.22	94.12±1.50	97.78±0.66	98.33±0.87	94.14±1.62	96.74±1.11	94.76±0.45	89.55±0.87	**99.70±0.04**
Kappa	0.7930±0.0861	0.9086±0.0448	0.9361±0.0211	0.9714±0.0107	0.9784±0.0150	0.9189±0.0268	0.9593±0.0118	0.9363±0.0035	0.8716±0.0082	**0.9962±0.0012**

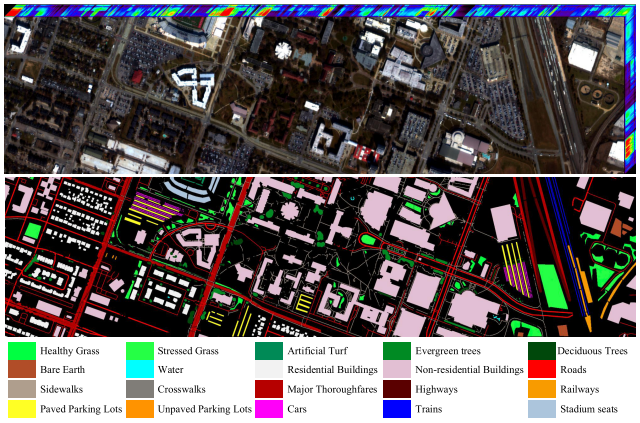

Fig. 14. False-color image and ground-truth map for the 2018 IEEE GRSS DFC dataset.

TABLE IX

NUMBERS OF TRAINING AND TEST SAMPLES FOR THE 2018 IEEE GRSS DFC DATASET

Class number	Class name	Training	Test	Total
1	Healthy grass	490	9759	9799
2	Stressed grass	1625	30877	32502
3	Artificial turf	34	650	684
4	Evergreen Trees	679	12909	13588
5	Deciduous Trees	252	4796	5048
6	Bare earth	226	4290	4516
7	Water	13	253	266
8	Residential buildings	1988	37774	39762
9	Non-residential buildings	11184	212500	223684
10	Road	2291	43519	45810
11	Sidewalk	1700	32302	34002
12	Crosswalk	76	1440	1516
13	Major Thoroughfares	2318	44040	46358
14	Highway	492	9357	9849
15	Railway	347	6590	6937
16	Paved Parking Lot	574	10901	11475
17	Unpaved parking Lot	10	139	149
18	Cars	329	6249	6578
19	Trains	285	5080	5365
20	Stadium Seats	341	6483	6824
	Total	25254	479908	504712

description of the classification maps. As shown in Table X, the quantitative results are presented in the form of mean value ± standard deviation by choosing random samples and calculating the result of ten times experiments. Our CLSJE outperforms the other methods in terms of OA, AA, and Kappa, and achieves the highest classification accuracy in most classes. Consequently, our algorithm yields the most satisfactory performance among all mentioned methods on the dataset with a low correlation between training samples and test samples.

V. DISCUSSION AND ANALYSIS

To better demonstrate the effectiveness of proposed components in the CLSJE learning network, we present ablation experiments to analyze the contribution of all components. Ablation experiments are conducted on the IP dataset, the HU dataset, and the 2018 IEEE GRSS DFC dataset (HU18).

The results are shown in Table XI. The IP dataset has low spatial resolution and imbalanced samples. The HU dataset has official training and test samples with high spatial resolution. In comparison, the 2018 IEEE GRSS DFC dataset has high spatial resolution and imbalanced samples. We chose this dataset for more comprehensive, reliable, and reproducible analysis results. For comparison, we select the baseline model, which is an encoder–decoder architecture (SegNet) trained by the hierarchically balanced (H-B) sampling strategy of SSDGL.

A. Discussion on the Cross-Level Feature Aggregation Framework

The CLA framework is described in Section III. We revised the baseline model and encouraged the cross-level feature flow. Significantly, the traditional coarse-to-fine flow and the fine-to-coarse feature aggregation are allowed. The classification performance of CLA without SFFM and SSJA by the H-B sampling strategy is shown in Table XI(b). Note that the SFFM

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TABLE X

CLASSIFICATION RESULTS ON THE 2018 IEEE GRSS DFC DATASET WITH 5% LABELED SAMPLES PER CLASS (BEST RESULTS ARE SHOWN IN BOLD)

Class number	CDCNN	FDSSC	SSFCN	SSRN	SSTN	DBMA	DBDA	FPGA	SSDGL	CLSJE
1	85.72±7.04	86.64±6.51	55.38±11.77	88.97±9.89	85.64±3.23	83.63±3.31	87.58±3.31	91.81±0.39	**93.40±0.07**	92.80±0.20
2	94.13±2.27	92.28±3.67	98.13±0.93	94.91±3.01	94.50±1.55	**95.11±1.12**	94.75±1.91	93.22±0.84	91.30±0.93	93.78±0.34
3	99.34±0.59	**100.00±0.00**	**100.00±0.00**	89.93±3.00	98.84±0.86	99.93±0.20	**100.00±0.00**	**100.00±0.00**	98.61±0.69	99.85±0.15
4	93.91±1.28	95.22±1.97	95.49±2.45	96.22±0.85	95.75±1.29	94.46±1.66	96.22±0.71	99.01±0.07	99.09±0.09	**99.58±0.04**
5	82.24±8.54	87.53±5.12	83.84±5.80	92.78±3.74	74.36±8.90	87.99±4.08	92.41±1.78	96.59±0.38	96.85±0.45	**97.30±0.20**
6	90.57±5.57	98.19±0.88	98.92±1.23	99.03±1.08	99.47±0.43	98.21±0.71	99.84±0.16	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
7	99.78±0.52	**100.00±0.00**	**100.00±0.00**	70.00±5.83	99.53±0.21	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
8	82.66±5.31	91.61±4.43	89.70±4.59	93.92±3.32	89.34±4.09	91.86±3.71	93.64±1.72	98.91±0.45	98.61±0.25	**99.36±0.20**
9	94.37±1.79	97.48±2.15	97.65±1.41	97.91±0.50	97.06±1.60	**97.79±0.62**	97.61±0.30	95.93±0.76	94.51±1.17	**97.79±0.05**
10	74.86±2.81	82.61±4.10	82.30±5.92	84.78±4.12	73.78±13.05	84.30±4.81	85.72±2.49	83.56±5.70	81.17±4.12	**93.55±1.02**
11	70.48±5.93	84.90±3.00	84.48±4.71	81.94±2.66	61.79±21.47	79.02±4.95	80.62±2.98	81.81±1.73	84.01±0.99	**87.52±3.05**
12	3.89±11.67	48.50±20.69	42.40±13.15	48.04±12.46	7.58±3.38	55.66±10.53	60.42±8.35	59.36±5.48	**65.12±1.56**	59.64±1.94
13	76.31±4.03	82.19±4.53	79.39±8.90	89.54±2.73	81.42±7.67	85.66±4.11	89.55±2.80	93.90±0.86	94.36±0.71	**96.84±0.62**
14	88.23±5.36	92.44±2.91	91.61±2.22	95.25±2.86	91.89±3.10	92.56±3.11	98.98±1.39	99.48±0.03	99.50±0.17	**99.83±0.07**
15	93.38±4.13	99.71±0.27	99.90±0.08	99.02±0.48	**99.91±0.11**	97.38±2.39	99.71±1.06	99.71±0.04	99.83±0.05	**99.91±0.01**
16	86.31±6.11	93.51±1.24	97.23±0.95	95.17±1.64	94.53±1.09	92.87±2.86	96.78±1.47	99.56±0.08	98.66±0.36	**99.80±0.07**
17	25.52±20.19	98.16±1.03	95.17±1.93	20.00±8.02	97.17±1.54	99.14±1.48	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
18	76.84±4.95	81.47±5.15	80.97±5.62	93.70±3.61	70.48±8.97	89.61±4.46	93.36±4.19	99.16±0.03	**99.32±0.23**	99.23±0.05
19	93.68±2.73	98.44±1.33	96.52±2.80	97.66±3.87	92.44±1.63	96.69±1.15	97.99±0.87	99.60±0.37	99.76±0.01	**99.84±0.08**
20	95.70±2.08	99.10±0.58	97.30±2.33	99.05±1.02	95.42±1.40	99.19±0.62	99.57±0.22	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
OA (%)	87.37±2.93	92.25±1.89	90.80±2.99	93.78±0.82	89.45±0.93	92.61±0.77	93.62±0.42	93.55±1.16	93.24±1.12	**96.64±0.91**
AA (%)	80.40±3.05	90.50±1.58	88.33±3.27	86.39±5.84	85.03±1.55	91.05±0.88	93.29±1.59	94.58±0.21	94.71±0.35	**95.83±0.26**
Kappa	0.8351±0.0218	0.8992±0.0162	0.8806±0.0140	0.9190±0.0107	0.8620±0.0127	0.9039±0.0099	0.9159±0.0055	0.9170±0.0143	0.9131±0.0131	**0.95665±0.0015**

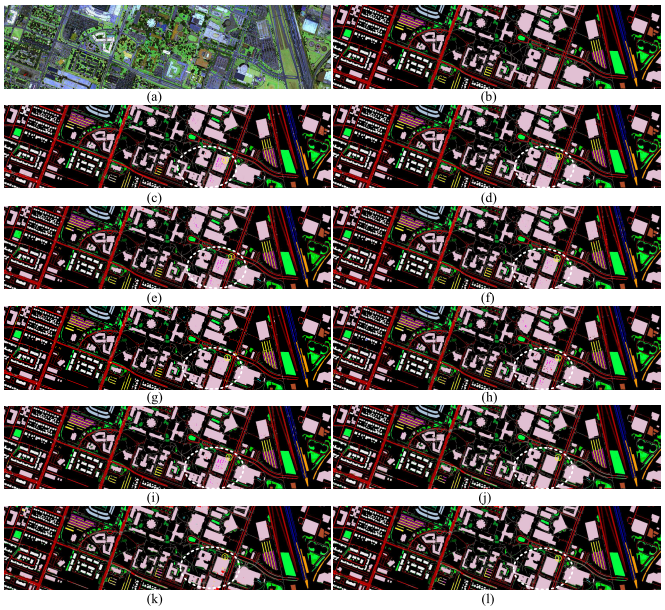

Fig. 15. Visualization of the classification maps for the 2018 IEEE GRSS DFC dataset. The white dashed lines outline part of the subtle classification distinctions. (a) False-color image. (b) Ground truth. (c) CDCNN. (d) FDSSC. (e) SSFCN. (f) SSRN. (g) SSTN. (h) DBMA. (i) DBDA. (j) FPGA. (k) SSDGL. (l) CLSJE.

is replaced only by a 1×1 convolution and attached addition operation. The results indicate that the CLA framework improves the OA, AA, and Kappa for the HU dataset (i.e., OA from 34.77% to 92.34%), the IP dataset (i.e., OA from 46.33% to 99.59%), and the 2018 IEEE GRSS DFC dataset (i.e., OA from 30.41% to 85.21%). The CLA framework is applied to introduce HSIs input pyramid in the encoder and aggregate multiscale features for the decoder. These results indicate that it is valuable to further exploit the multiscale spatial features and cross-level features for HSI classification.

B. Discussion on the Supervised Feature Fusion Module

Table XI(d) represents the classification performance of the CLA framework adding SFFM. There are substantial drops in terms of OA, AA, and Kappa on the HU dataset and lightly drops on the IP dataset and the 2018 IEEE GRSS DFC dataset. The reason is that Table XI(d) incorporates SFFM makes the network deeper and introduces numerous redundancy high-level features. Numerous redundancy features increase the difficulty in extracting discriminative features. However, the encoder layer of Table XI(d) only consists of a 3×3 convolutional layer, resulting in the dropped classification accuracy of Table XI(d). The quantitative results of Table XI(h) demonstrate that SFFM and SSJA together achieve significant gain. With the help of SSJA, the network focuses on discriminative information and bypasses redundant information to boost classification performance. These results show that the SFFM combined with a strong encoder layer improves classification performance. To further confirm the reason, we replace a single convolutional layer with the different architectures of encoder layers to demonstrate that the SFFM combined with a strong encoder layer is efficient for classification performance.

We replace a single convolutional layer with multiple convolutional layers [Variant (b)], encoder layer of FPGA [Variant (d)], and encoder layer of SSDGL [Variant (f)]. The quantitative results are presented in Table XII. Variant (c), Variant (e) and Variant (g) denote the Variant (b), Variant (d) and Variant (f) without the SFFM component, respectively. As shown in Table XII, the classification accuracy is gradually improved by adding attention mechanisms and increasing network depth. Although incorporating SFFM increases the numerous features, the single convolutional layer cannot leverage these features efficiently and even result in classification drops. The quantitative results confirm that SFFM needs to be effectively exploited combined with extra attention mechanisms or a deeper convolution network. The results of Variant (c), Variant (e), and Variant (g) demonstrate that

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TABLE XI

HSI CLASSIFICATION RESULTS EVALUATED ON THE HU DATASET AND THE IP DATASET WITH TEN LABELED SAMPLES PER CLASS AND 5% LABELED SAMPLES, RESPECTIVELY. THE ENCODER–DECODER BASELINE, THE CLA, SFFM, SSJA, AND CSS ARE ADDED IN CLSJE FOR THE MODULE ANALYSIS

Method	CLA	SFFM	SSJA	CSS	OA (%)			AA (%)			Kappa		
					HU	HU18	IP	HU	HU18	IP	HU	HU18	IP
(a) Baseline	-	-	-	-	34.77 ±4.56	30.41 ±6.22	46.33 ±3.89	41.23 ±3.90	36.92 ±6.02	64.04 ±3.13	0.2974 ±0.0254	0.2536 ±0.0377	0.4146 ±0.0181
(b) CLSJE w/o SFFM, SSJA and CSS	✓	-	-	-	92.34 ±1.61	85.21 ±2.82	99.59 ±1.02	93.57 ±1.39	83.65 ±2.31	99.80 ±0.96	0.9172 ±0.0174	0.8284 ±0.0218	0.9953 ±0.0122
(c) CLSJE w/o SFFM, SSJA	✓	-	-	✓	94.94 ±0.71	86.97 ±0.96	99.63 ±0.64	95.65 ±1.42	85.71 ±2.01	99.82 ±0.49	0.9452 ±0.0080	0.8421 ±0.0124	0.9957 ±0.0062
(d) CLSJE w/o SSJA and CSS	✓	✓	-	-	49.09 ±5.20	73.45 ±3.89	98.76 ±1.22	53.63 ±4.26	70.11 ±3.53	99.29 ±0.67	0.4514 ±0.0329	0.6729 ±0.0289	0.9858 ±0.0082
(e) CLSJE w/o SSJA	✓	✓	-	✓	57.23 ±4.19	80.21 ±3.08	98.85 ±1.10	65.13 ±3.71	76.67 ±3.01	99.37 ±0.51	0.5405 ±0.0297	0.7333 ±0.0210	0.9870 ±0.0067
(f) CLSJE w/o SFFM and CSS	✓	-	✓	-	99.82 ±0.58	95.37 ±1.67	99.77 ±0.40	99.84 ±1.03	94.42 ±1.25	99.86 ±0.57	0.9980 ±0.0208	0.9268 ±0.0220	0.9974 ±0.0056
(g) CLSJE w/o SFFM	✓	-	✓	✓	99.87 ±0.44	95.72 ±1.46	99.85 ±0.27	99.88 ±0.55	95.13 ±1.05	99.93 ±0.23	0.9986 ±0.0106	0.9389 ±0.079	0.9986 ±0.0048
(h) CLSJE w/o CSS	✓	✓	✓	-	99.92 ±0.07	96.28 ±1.06	99.83 ±0.02	99.94 ±0.05	95.40 ±0.21	99.91 ±0.02	0.9992 ±0.0008	0.9522 ±0.029	0.9981 ±0.0002
(i) CLSJE	✓	✓	✓	✓	99.95 ±0.03	96.64 ±0.91	99.99 ±0.01	99.96 ±0.02	95.83 ±0.26	99.99 ±0.00	0.9995 ±0.0002	0.9566 ±0.0015	0.9998 ±0.0001

TABLE XII

ABLATION STUDY ON SFFM. QUANTITATIVE COMPARISONS AMONG VARIANTS OVER THE HU DATASET AND THE 2018 IEEE GRSS DFC DATASET

Variants	OA (%)		AA (%)		Kappa	
	HU	HU18	HU	HU18	HU	HU18
(a) CLSJE w/o SSJA and CSS	49.09	73.45	53.63	70.11	0.4514	0.6729
(b)	62.21	75.29	57.47	72.68	0.5437	0.7198
(c)	92.12	85.64	93.06	83.89	0.9154	0.8321
(d)	99.54	94.65	99.63	93.78	0.9950	0.9265
(e)	98.89	93.43	99.22	93.12	0.9887	0.9211
(f)	99.77	95.34	99.81	94.69	0.9975	0.9302
(g)	98.52	93.95	98.62	93.66	0.9769	0.9253
(h) CLSJE w/o CSS	99.92	96.28	99.94	95.40	0.9992	0.9522

TABLE XIII

ABLATION STUDY ON SSJA. QUANTITATIVE COMPARISONS AMONG VARIANTS OVER THE HU DATASET AND THE 2018 IEEE GRSS DFC DATASET

Variants	OA (%)		AA (%)		Kappa	
	HU	HU18	HU	HU18	HU	HU18
(a) CLSJE w/o HFSA	98.98	94.88	98.98	95.47	0.9899	0.9339
(b) CLSJE w/o STCA	97.89	93.11	97.91	93.32	0.9789	0.9269
CLSJE	99.95	96.64	99.96	95.83	0.9995	0.9567

SFFM is effective in most architectures of the encoder layer. Furthermore, Table XII(d) and (f) demonstrates that introducing attention mechanisms gets more signification improvement than increasing the network depth.

C. Discussion on the Spectral–Spatial Joint Attention Module

Compared to Table XI(b) and (c), the results of Table XI(f) and (g) show that incorporating SSJA achieves significant gains. SSJA consists of HFSA and STCA, which, respectively, reweights the feature maps in the spatial and spectral streams and models the spectral–spatial representation, which extracts discriminative HF features. We conducted an ablation experiment on the HU dataset and the 2018 IEEE GRSS DFC dataset to analyze the contribution of two attention components.

As shown in Table XIII, the quantitative results of Table XIII(a) and (b) demonstrate that the proposed HFSA and STCA effectively extract discriminative features and obtain the expected improvement of classification accuracy. The quantitative results of our CLSJE show that combining the HFSA and STCA improves the learning effectiveness of the HF information in spatial and spectral dimensions, and results in higher classification accuracy.

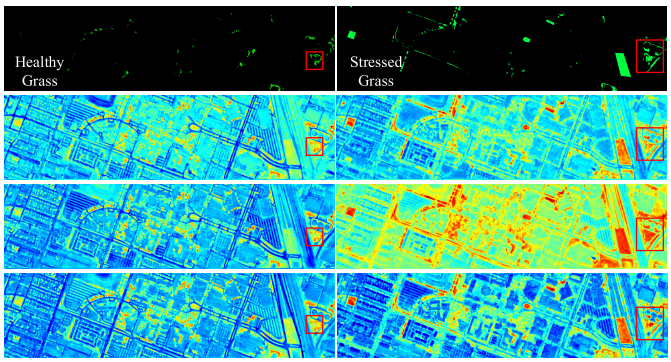

Fig. 16. Feature maps for "Healthy Grass" and "Stressed Grass" on the 2018 IEEE GRSS DFC dataset. From (Top) to (Bottom) represent ground truth, Variant (a), Variant (b), and our CLSJE.

As shown in Fig. 16, we visualize the feature maps of the last layer of variants and our CLSJE. In Fig. 16, we display the feature maps of "Healthy Grass" and "Stressed Grass." Variant (b) without HFSA shows poor performance in extracting the discriminative HF feature and is difficult to distinguish the "Stressed Grass" and "Healthy Grass." Although Variant (a) with HFSA has a signification performance in distinguishing the "Stressed Grass" and "Healthy

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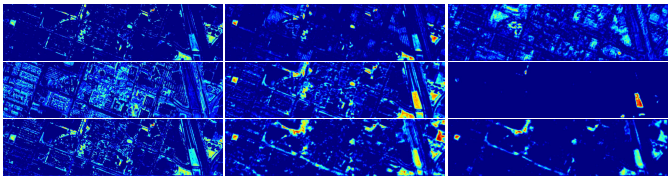

Fig. 17. Grad-CAM visualization results for "Stressed Grass" on the 2018 IEEE GRSS DFC dataset. From (Left) to (Right) represented that the grad-CAM visualization is calculated for the first, second, and third encoded layer outputs. From (Top) to (Bottom) represent ground truth, Variant (a), Variant (b), and our CLSJE.

Grass," Variant (a) without STCA exhibits low confidence in the object and less distinction between object and context. The feature maps of our CLSJE confirm that the exploiting of HFSA and STCA can capture fine-grained detail and results in pleasant performance in fine-grained classification. The activation areas captured by the model tend to focus on the objects, making the features more structural, and redundant information is significantly reduced.

By observing the regions that the network has considered important for predicting a class, we attempt to look at how this network is making good use of features. We compare the Grad-CAM visualization results of Variant (a), Variant (b), and our CLSJE. Fig. 17 illustrates the visualization results corresponding to the class "Stressed Grass." Qualitatively, the Grad-CAM of our CLSJE almost overlaps with the object regions. That is, our network with HFSA and STCA learns well to exploit information in target object regions and aggregate features from them.

D. Discussion on the CSS

Table XI(f) presents the results of the CLSJE with the CSS. For the IP dataset possessing imbalanced samples, CSS results in an OA improvement from 99.83% to 99.99% and Kappa improvement from 0.9981 to 0.9998. The OA is improved from 96.28% to 96.64% and AA from 95.40% to 95.83% on the 2018 IEEE GRSS DFC dataset. These quantitative results suggest the CSS assists in classifying HSIs, especially for the imbalanced sample classification task, with substantial improvement. The training samples of each class are selected by proportion and inputted into the hierarchical sequence in random order, which can keep the combination stochasticity and obtain stable yet diverse gradients. The OA is improved from 99.92% to 99.95% and AA from 99.94% to 99.96% on the HU dataset. The HU dataset uses a fixed number of samples per class. These quantitative results show that only using an inverse-weighted loss strategy improves classification accuracy. Besides, through the inverse-weighted loss strategy, the model convergence is further sped, the easily misclassified classes are emphasized, and the classification accuracy is further improved. Meanwhile, introducing CSS obtains an improvement in classification accuracy for all variants.

E. Discussion on the Running Time

Table XIV lists the training and testing time of the six methods on the IP, PU, and HU datasets. Compared to the patch-based HSI classification methods (i.e., SSRN, FDSSC, SSTN, and DBDA), the patch-free techniques (i.e., FPGA, SSDGL, and CLSJE) consume less computing time and achieve desired performance. The patch-free methods process the whole image rather than a patch of each test pixel, so the inference speed is greatly improved due to the reduction of the overlapping areas between the adjacent pixel patches. Less testing time is essential to embed in satellite and uncrewed aerial vehicle (UAV) platforms.

TABLE XIV
TRAINING AND TESTING TIMES OF DIFFERENT METHODS ON THREE HSI DATASETS

		IP	PU	HU
FDSSC	Train (min)	78.0	59.0	92.9
	Test (s)	4.6	13.2	5.8
SSRN	Train (min)	21.9	27.3	34.6
	Test (s)	1.7	3.8	5.9
SSTN	Train (min)	13.8	38.3	36.3
	Test (s)	1.1	3.4	3.7
DBDA	Train (min)	30.0	21.5	55.3
	Test (s)	5.4	14.7	6.6
FPGA	Train (min)	0.97	1.79	9.3
	Test (s)	0.25	0.29	0.20
SSDGL	Train (min)	1.84	5.57	23.2
	Test (s)	0.24	0.41	0.40
CLSJE	Train (min)	1.4	3.7	15.9
	Test (s)	0.23	0.37	0.36

DBDA, FDSSC, and SSRN exploit deep residual networks to extract spectral and spatial features, requiring much training and testing time. SSTN, as the factorized architecture search framework, aims to design the optimal deep residual architecture. It obtains better classification performance with sacrificing speed. FPGA, SSDGL, and our method are FCN-based methods. These methods significantly improve the speed of training. The training time and the inference time of our CLSJE are slightly longer than those of FPGA, but the classification accuracy achieves a significant improvement. Compared to SSDGL, our method obtains a tradeoff between speed and accuracy.

VI. CONCLUSION

In this article, we present a CLSJE framework for HSI classification tasks. In contrast to existing patch-free HSI classification models, our method improves the classification performance of HSIs with imbalanced samples further in terms of speed and accuracy. Our CLSJE employs CLA architecture, which performs multiscale inputs and multiple-to-one connections instead of traditional single input and lateral connections. These designs significantly reduce the difficulty of training, enabling the proposed framework to capture both the global semantics information and the local context in multiscale views. Especially, the SSJA mechanism consisting of the HFSA and STCA is proposed to exploit the interdependence between spectral features. Particularly, SSJA can be seamlessly integrated into different architectures for HSI tasks. A CSS is exploited to accelerate convergence speed and alleviate the issue of insufficient or imbalanced training samples. Evaluations of four public HSI datasets confirm that CLSJE achieves state-of-the-art classification performance in visible and quantitative results.

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In the future, we will further improve the ability of spectral–spatial feature learning and seek a balance between speed and precision, which will be fundamental for the real-time classification of HSIs from satellite and UAV platforms. We hope that the proposed modules for spectral–spatial feature learning will aid future research in HSI classification and other HSI tasks.

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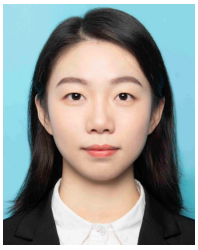

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